1. The Issue with Urban Heat Islands

Urban heat islands (UHIs) are land areas with elevated levels of heat as a result of the built environment and population density, where temperatures can differ by more than 10 degrees compared to nearby rural areas due to factors such as concrete surfaces that absorb and re-emit heat, tall buildings that trap heat, and loss of trees and natural areas (Climate Central, 2025; Durham County, n.d.). Building on this, a research paper by Chanpichaigosol et al. (2025) comes to a similar conclusion, explaining that UHIs are characterized by temperatures higher than their surroundings, and that this process is driven largely by urbanization that leads impervious surfaces to absorb heat whenever possible and release it back into the environment regardless of time of day.

The phenomenon occurs specifically in urban environments because they have:

a. More dark surfaces and concrete that hold heat
b. Fewer trees to provide shade and evapotranspiration
c. Dense building configurations that trap heat
d. Reduced natural cooling mechanisms compared to rural areas

Here are crucial opening questions surrounding the issue of Urban Heat Islands.

How do materials in the urban environment (built/construction, vegetation/greenspace) impact temperatures?

Different materials in cities have opposite effects on temperature, with building materials making areas hotter while plants and trees help cool them down. Concrete, asphalt, and building surfaces act as heat collectors because they absorb and hold the heat from sunlight instead of reflecting it away, creating surfaces that, as Chanpichaigosol et al. (2025) notes, absorb heat during daytime and release it slowly over time, including at night. This same article includes research from Chiang Mai showing a strong connection between areas with lots of buildings and higher temperatures, where neighborhoods with many buildings store heat energy effectively, while tall buildings make the problem worse by trapping hot air and blocking wind that would normally provide cooling. Durham County (n.d.) came to the same conclusion. Climate Central's 2025 report makes a similar point by explaining that roads, sidewalks, parking lots, and building facades take in far more solar energy than natural surfaces and radiate it back into the city as heat. Trees and plants work differently, cooling areas through a process where plants absorb heat and carbon dioxide while releasing water vapor that lowers air temperatures around them, plus they provide shade that blocks direct sunlight and allow rainwater to soak into the ground rather than run off hot pavement. Both Durham County (n.d.) and Chanpichaigosol et al. (2025) describe how this combination of shading, moisture release, and improved water absorption helps cool nearby surfaces and reduces overall temperatures.

Chanpichaigosol et al. and Climate Central both show that areas with more vegetation consistently experience lower temperatures because plants cool the air through this natural process. Materials like concrete effectively work as heat storage systems that collect energy during the day and release it slowly over many hours, keeping temperatures high even after sunset, while plants use water from the soil to create cooling effects through natural processes that concrete cannot copy. Research summarized by Chanpichaigosol et al. (2025) shows that strategically placing trees and vegetation can reduce urban heat island effects by as much as 64 percent in areas with lots of buildings. These differences between materials show that choosing the right materials when designing cities directly affects how comfortable people feel and is important for creating livable urban areas. Beyond the impacts of materials on urban heat, these landcovers also spatially vary across a city. A study looking at the spatial-temporal patterns of urban heat islands in China, found that the development of localized urban centers spread throughout a major city resulted in a distribution of urban heat, which spiked in these areas where there was intensive development (Yang et. al, 2025). This shows that not only the types of landcovers (impervious or otherwise), and their resulting quantities, impact urban heat, but so do their spatial placement within the city. We aim to test this by increasing and decreasing the clustering of landcovers to see if it leads to any different outcomes.

Why is urban heat important overall?

Urban heat is important because it creates serious health risks and worsens existing inequalities in cities worldwide. When urban areas become significantly hotter than surrounding rural areas due to concrete and buildings absorbing heat, research from Durham County (n.d.) and Chanpichaigosol et al. (2020) shows that it leads to dangerous health problems including heat exhaustion, heat stroke, respiratory difficulties, and even death, particularly affecting vulnerable populations like elderly people, children, pregnant women, and those with existing health conditions. The problem disproportionately impacts low-income communities and historically marginalized neighborhoods that often lack adequate green spaces, air conditioning, and other resources to combat extreme heat, perpetuating social inequalities that stem from past discriminatory policies like redlining. Additionally, urban heat creates a damaging cycle where increased temperatures force greater use of air conditioning, leading to higher energy consumption and greenhouse gas emissions that further worsen both climate change and urban heating. Understanding and addressing urban heat is therefore critical for protecting public health, promoting social equity, and creating livable cities as urbanization continues to expand globally.

What makes urban heat complex?

When thinking about what makes urban heat complex, different materials such as grass, concrete, and trees and the impacts that can have on the diffusion of heat across an urban system play into complexity. These materials at a micro level influence heat (the properties interacting with temperature), but also at a macro level, their placement amongst each other in the wider layout of the city impacts temperature and how it diffuses across landcovers. This is where using ABM as a tool to understand urban heat is useful. ABMs allow simulation of individual entities (patches and agents) following simple heating, cooling, and diffusion rules that collectively produce emergent temperature fields. As Yang et al. (2016) highlights, the composition of urban environments leads to heat transfer that is uneven, and ABM is useful then for understanding the micro-level interactions that can lead to this uneven heat transfer. In this case, it's the micro-level interactions between land-cover elements (patches) and vegetative agents (trees), enabling exploration of both quantitative (percent coverage) and spatial (layout and placement) effects on heat diffusion

Oke (1982) describes how cities are inherently complex and the effects they have on the atmosphere are similarly complex. This is because the atmospheric state responds to exchanges of energy, mass, and momentum, and the sources and sinks for these exchanges are very heterogeneous and respond to both natural and anthropogenic factors. Next, Oke how difficult it is to observe these patterns in urban areas. There is a lot of space to measure, it's difficult to measure over time without some sort of disruption to the environment. Lastly, it's expensive. The costs of all the equipment, plus the time needed to monitor and maintain it, add up quickly. When you combine that with the spatial interaction effects of various land cover types , it becomes clear that urban heat is an incredibly complex issue. In order to understand how urban heat is manipulated by landcovers, it is important to tease out the more micro interactions that they have with one another.

2. Our Research Question

How do different spatial distributions and quantities of various land covers impact heat interactions in an urban environment?

Urban Heat Islands (UHIs) emerge from interacting material, geometric, and biological factors. Yang et al. (2016) explains that the UHI effect is closely related to urban heat release, which is determined by many different factors such as vegetation coverage, population density, weather conditions, and the underlying surfaces. These multiple factors, and their relationships, create temperature patterns that cannot be fully captured using a simple mean, and therefore a spatially explicit model using ABM is needed to best represent the relationship between all of the different factors.

Looking at the different factors more closely, Druckenmiller (2023) explains that impervious surfaces like concrete absorb more heat than natural surfaces such as vegetation. Additionally, Yuan et al. (2025) explains that UHI intensity is higher in areas with more impervious surfaces, which supports the need for our model to help represent how these different percentages of landcovers have a tangible effect on the outcome of our model, and why it's important to research.

More research has further supported this, with Yang et al. (2016) noting that vegetated outskirts (rural/suburban areas) and urban cores can have a temperature difference range between 3°C and 8°C. Xu et al. (2025) also notes that it's the reduction of impervious surfaces that helps mitigate UHIs, showing how it's the amount of each landcover-type that determines the magnitude of UHI intensity, with more impervious surfaces meaning increasingly severe UHIs.

3. The Model (Components & Order of Events)

3.1 Model Components

There are a couple of agents that are fundamental to the urban heat island effect (UHIE) and thus need to be included in the model. Impervious surfaces made of asphalt (roads, sidewalks, parking lots, etc), trees, and grass (broadly includes non-shade plants). Impervious surfaces are well-documented and known to be the main contributors to the UHIE, and models of the problem cannot exist without the inclusion of them. On the other hand, grass and trees need to be included because they are the main alleviators of the UHIE. Areas can have lots of asphalt, but very high shade from trees, and therefore very little UHIE. Grass, while it does not provide shade, absorbs the heat and does not release it back while asphalt reflects and stores heat, increasing ambient temperature. We also made sure to implement temperature as sliders so that we can mimic daily temperature curves in different cities known to be facing urban heat, thus making our model more generalizable.

3.1.1 Excluded Components

We thought about other agents such as the use of air conditioning, appliances, buildings, and vehicular traffic on the roads, as these are other factors that are known to contribute to the UHIE through pollution and other means. However, there would need to be a high degree of assumption about the extent of the effects that each one of these individually has, and the effects overall from each are more or less the same, so to avoid unnecessary complication, we thought it best to omit them from our model.

3.1.2 Patch Descriptions

First, there is the patch-related variable of airtemp, which is the temperature of the patch at a given time and changes based on the time of day, landcover, shade from trees, and diffusion of heat. Next there is the patch-related variable of landcover, in which a random numeric value is assigned to each patch and later used to determine the landcover-type. There is also a patch-related variable called landcover-type. This uses the random value from landcover, and patches are either designated as concrete or grass. Lastly, there is a patch-related variable called has-tree and a patch variable called near-tree. The former takes on a boolean value, representing whether a patch has a tree on it or not, while the latter stores if a patch is within a radius of at least one tree.

3.1.3 Global Variables

There are various global variables that are used to both calibrate the model and use as parameters that can create and test customized layouts/environments.

3.1.3.1 Scenario Parameters

A concrete slider is used to determine the quantity of the concrete landcover, with the remaining amount being the grass landcover. In order to test spatial density, a clustering slide is used to determine how densely populated the landcovers will be. To test the impacts of trees, the num-trees slider and the tree-radius slider determine the number of trees present and how many patches they impact with their cooling effect respectively.

3.1.3.2 Generalizable Parameters

In order for the model to be generalizable to various cities with varying climates, there are several parameters that can be manipulated to represent the place or situation one wants to model. The random-layout? switch influences whether the landcovers will be placed randomly throughout the environment or whether it will be 100% clustered, and split according to quantity. The strategic-trees switch determines if trees will be located on grass patches near concrete patches or if they will be randomly placed on grass patches. The sliders for minimum temperature, maximum temperature, and the decline of temperature can be manipulated to set the base temperature of a given area, which will be manipulated by the flow of heat through the landcovers.

3.1.3.3 Calibration Parameters

Given our model components, specifically the flexible nature of the global variable sliders, in order for the model to behave in accordance with the basic scientific understanding of how different materials exhibit temperature , parameterization needs to occur. The following variables are parameterized:

- Inflow-rate: Determines how much of the base temperature enters the system
- %-inflow-removed: How much heat is removed from the base temperature as it goes into patches that are impacted by the shading of trees
- outflow-grass: How much heat leaves grass patches
- outflow-concrete: How much heat leaves concrete patches
- tree-outflow: A cooling amplifier for how much heat leaves patches impacted by the shading of trees
- diffuse-slider: how much of a landcovers air temperature is passed to its neighbors

In order to guide parameterization, a study conducted in the UK, testing the impacts of trees and grass on urban surface temperatures, was used to both theoretically and empirically guide the inflow and outflow related parameters. The study found that the surface temperature of concrete was around 17 degrees Celsius higher than peak air temperature in the sun, and around 4 degrees higher than peak air temperature in the shade. Grass, due to its ability to perform evapotranspiration, had a surface temperature that was around 1 degree Celsius cooler than the peak air temperature in the sun, and around 4 degrees cooler than the peak air temperature in the shade (Armson, et al., 2012). These findings were used as rough estimates for what direction of temperature change concrete and grass should behave when in sun vs. in shade in our model.

Inflow/Outflow Parameters

In the most basic version of the model, inflow-rate will be 1.0 as 100% of the heat will enter the system, where . When playing with the tree-inflow, we understand that trees are on top of grass patches, so their role is to further amplify cooling the grass patch they are on, as well as the patches within their radius, i.e. shading effects. %-inflow-removed will be rather low (0.05) as it will block a little bit of heat coming in (5% of the base temperature will be blocked), but the sun's radiation can still penetrate through gaps in the foliage. Using the temperature differences the study found as rough estimates, we began to manipulate the inflows and outflows, while also using the basic theoretical concept that grass outflow should be higher than that of concrete outflow, as concrete holds on to heat more strongly, due to its material properties. After testing various values, we settled on an outflow-grass value of 0.50 and an outflow-concrete value of 0.43, which gave us overall temperature results that reflected the differences in temperature the article presented. We know that trees will have a significant cooling effect, through evapotranspiration, a value of 0.20 for tree-outflow was used as an amplifier for cooling, on top of the cooling that the grass patches already exhibit. The literature by Armson et al. (2012) explains that surface temperatures of grass in shade can be 4–7 °C cooler, so in our testing we found that a tree-outflow of 0.20 exhibited this difference when compared to grass patches not found in shade.

Beyond inflow/outflow rates, the diffuse-slider is also a parameter that is a part of the basic mechanism of urban heat. We needed to balance the smoothing of air temperature across the surface of the model while still preserving the various differences in air temperature created by the landcovers. In our testing, we found that having the patches diffuse 10% of their temperature outwards to neighboring patches allowed both of those conditions to be true.

The default settings for the parameters are as follows:

inflow-rate	1.0
tree-inflow	0.05
outflow-grass	0.50
outflow-concrete	0.43
tree-outflow	0.20
diffuse-slider	0.1

3.2 Order of Events

3.2.1 The Setup Procedure

3.2.1.1 Setting up the landcover

The model first sets up the landcovers in the environment. Each patch is assigned a landcover type either randomly or by coordinates, which is determined by the random-layout switch. If the layout is set to random, then the spatial density of the landcover types are applied to the environment via the clustering slider. The quantity of each landcover type is determined by the concrete slider, which represents what percentage of the landscape is concrete. If the random layout is turned off, a coordinate split is applied to the environment, and only the quantity of each landover type (via the concrete slider) is used to apply the landcovers. Once the landcovers are set up, the patches obtain an attribute called airtemp, which is initially set to the patch variable “temp-inflow”, which represents the temperature that is coming into the patches. After the landcovers are set, the patches are colored. Concrete patches are gray, and grass patches are green.

3.2.1.2 Setting up the trees

Now the trees are placed in the environment. Depending on whether or not strategic tree placement is selected, the number of grass patches that are eligible to have a tree on it will change. If the strategic trees option is turned on, then grass patches that are within 5 patches of a concrete patch will be labeled as eligible to have a tree on it. If the strategic option is not chosen, then all grass patches are labeled as eligible to have a tree on it. A number of the newly labeled eligible grass patches, determined by the num-trees slider, are then randomly selected, and colored a dark green, indicating that the grass patch now has a tree on it. Lastly, any patches within a determined radius away from the tree will be noted to be near a tree, this will determine the patches that will be impacted by the trees cooling abilities.

3.2.2 The Go Procedure

Now that the entire landscape is set, the landcovers and the trees, the temperature will enter the system iteratively, and the model environment will be colored in accordance with the air temperature present at each patch.

3.2.2.1 Base Temperature (updating the initial temperature inflow)

In order to set up the base temperature, we have derived a temperature curve using the maximum temperature, minimum temperature, and temperature decline slider values. This temperature curve resets every 24 ticks, each tick representing an hour, mimicking a day cycle.

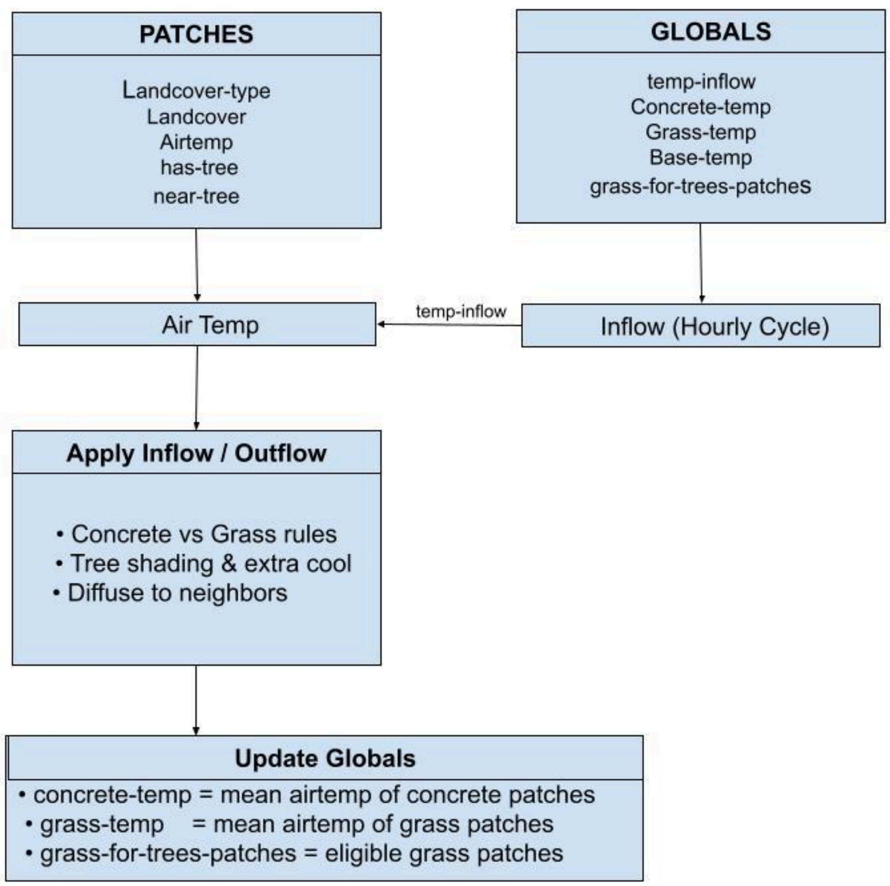
3.2.2.2 Temperature Inflow

The base temperature, based on the time of day, is multiplied by the inflow rate slider, to determine how much heat is able to enter the patch. For patches with trees, the %-inflow-removed slider is referenced, and indicates how much heat is unable to enter patches that are impacted by trees and their radius.The air temp of the patches is then set to include the inflow rate of heat.

3.2.2.3 Temperature Outflow

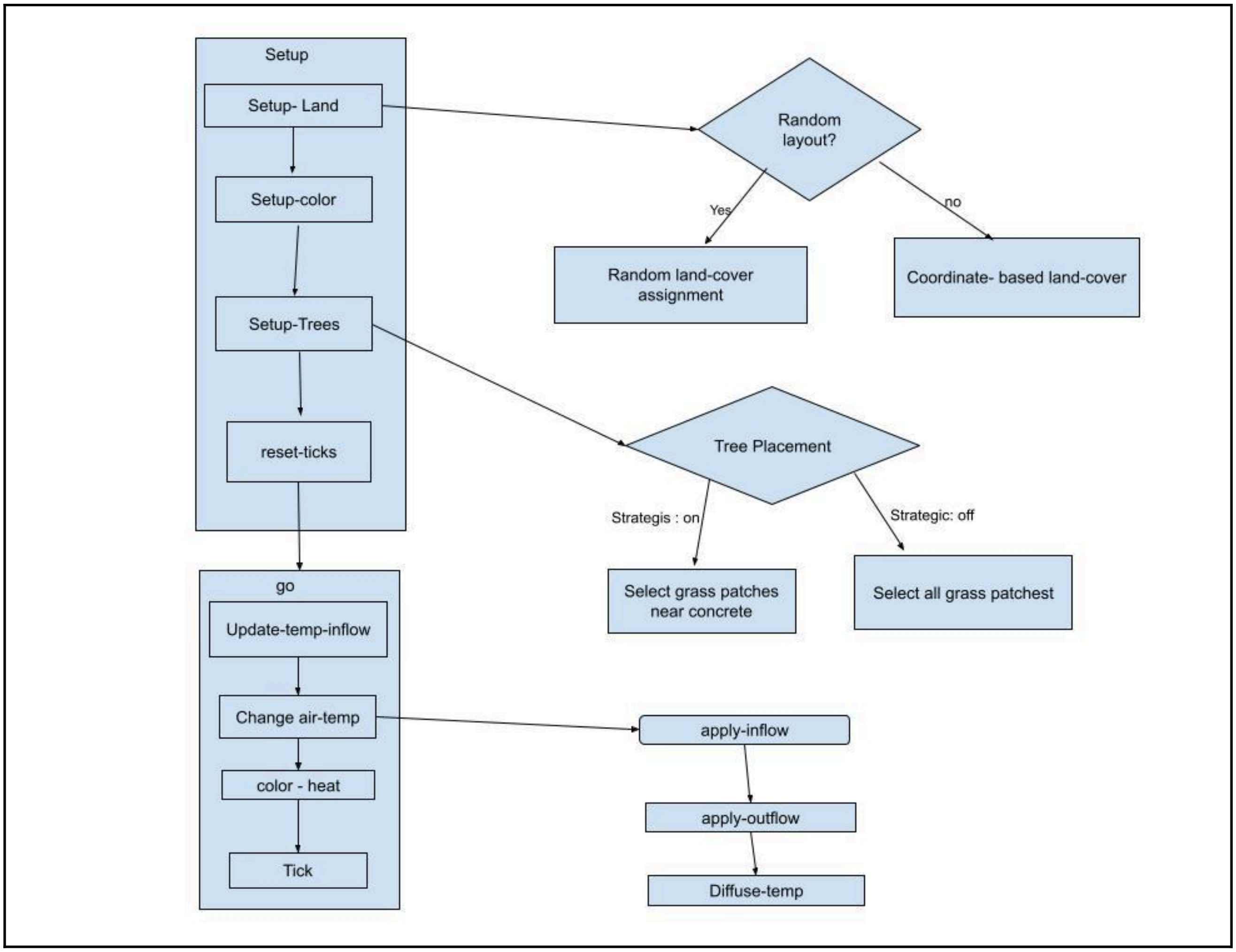
Now that the patches have a temperature, they then release heat. This is based on landcover type, as theoretically different materials release heat/hold on to heat at different rates. The amount of heat each patch lets go is determined by their corresponding slider: “outflow-concrete”, “outflow-grass”, and “outflow-trees”. The airtemp of the patches now subtract away the outflow rate, leaving the patches with a temperature that combines the heat they are receiving and the heat they get rid of. This new air temp is diffused out to neighboring patches, the rate of diffusion is determined by the “diffuse-slider”. Finally, the patches are colored on a scale of red, with cooler temperatures showing a lighter shade and hotter temperatures showing a darker shade.

Improved Conceptual Model Diagram



3.2 Order or Events

Order of Events Diagram



3.3 Sensitivity Testing

Metric	Clustering 0	Clustering 100	Clustering 500	Clustering 1000
Overall Mean Temperature	**82.885**	**83.226**	**83.243**	**83.241**
Grass	**75.97**	**71.866**	**71.715**	**71.763**
Concrete	**89.809**	**94.586**	**94.771**	**94.991**

As we were testing each of the scenario parameters, we found that clustering had a significant effect on temperature. We ran clustering at various values of 0, 100, 500, & 1000. When looking at the overall mean temperature, the mean temperature of grass, and the mean temperature of concrete, we found little differences in the temperature between 100 and 1000. This means that clustering effects change the temperature at smaller clustering values. After further testing, we found that the max clustering should be 10 for our scenario testing, as the differences in temperature became much smaller as clustering increased.

4. Scenario Simulations and Results

We will be varying:

a. The percent of each land-cover type

b. The spatial arrangement

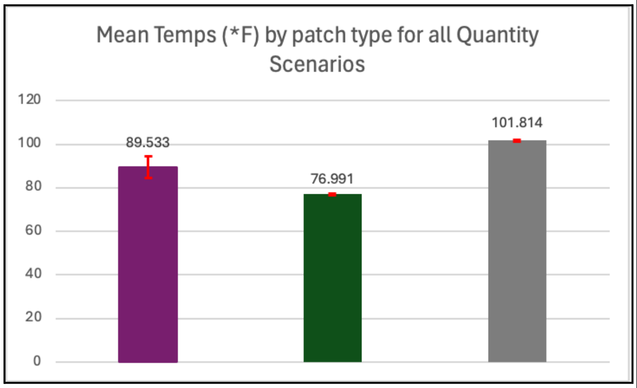
c. The quantity and radius of trees

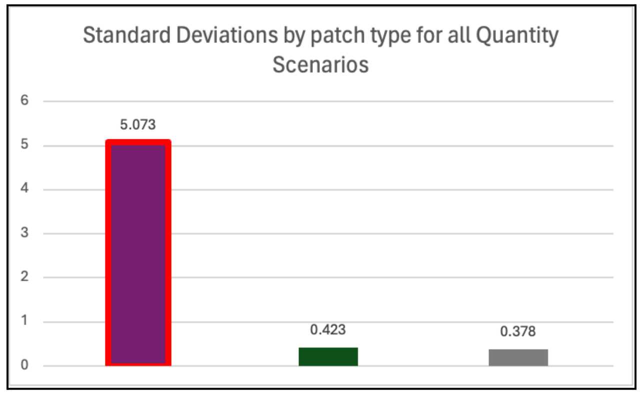
Thinking about our research question and what scenarios would help us explore the quantity and spatial distribution, and how these affect temperatures, we first thought of testing the Concrete slider, which controls the quantity of concrete and grass in our model.

Concrete/Quantity Scenario Testing

Changing Parameters			
Concrete	Clustering	Number Trees	Radius
25/50/75	10	10	4

Boston	Overall Mean (30 runs each)	Grass Mean (30 runs each)	Concrete Mean (30 runs each)
25% Concrete	**83.335**	77.31	101.408
50% Concrete	89.503	77.145	101.86
75% Concrete	**95.761**	76.517	102.174



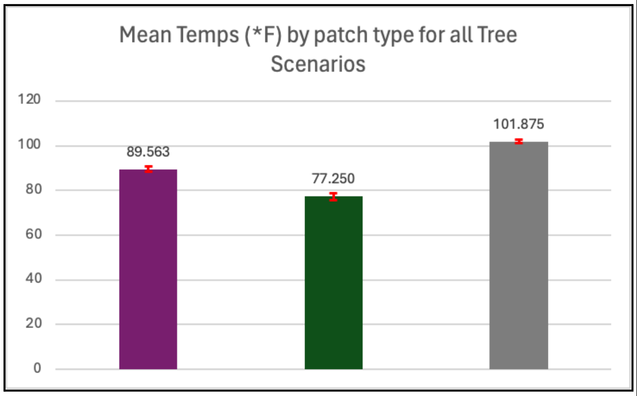


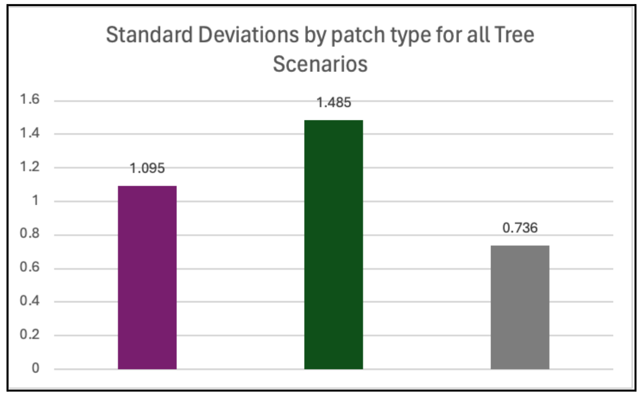
These results confirm existing scholarly research showing that simple quantities of concrete and grass are major drivers of temperature change in urban areas. This is because the overall temperature increased by 12.426*F when shifting the quantity of concrete from 25% of our model to 75%. Something that is worth noting though is that the means of all grass patches and all concrete patches on their own stayed relatively constant regardless of quantity. These differences in how much the temperature of the overall model, the grass patches, and the concrete patches can also be seen in the standard deviations, with the overall means having a much higher standard deviation than other patch types.

Tree Scenario Testing

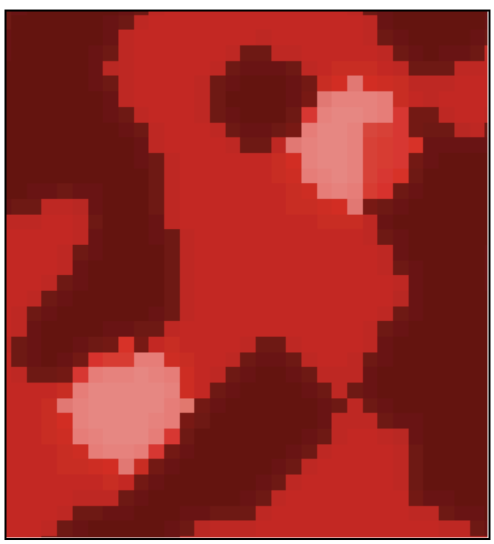
Changing Parameters			
Concrete	Clustering	Number Trees	Radius
50	10	0/10/20	4

Boston	Overall Mean (30 Runs each)	Grass Mean (30 runs each)	Concrete Mean (30 runs each)
0 Trees	90.928	79.108	102.745
10 Trees	89.503	77.145	101.86
20 Trees	88.258	75.498	101.019





Looking at how our model was affected by adding in the trees, the results confirm existing research that trees provide shading that lowers temperatures. However, there is a greater standard deviation on the grass patches, which can be explained partly by how our model is coded. This was something we should have changed sooner, but we kept strategic trees on for all of our scenarios. What this means is that trees can only spawn within 5 patches of concrete, but since their radius was set to 4 for testing, most of the trees' shading appeared on grass patches. See the picture below to understand what I mean.

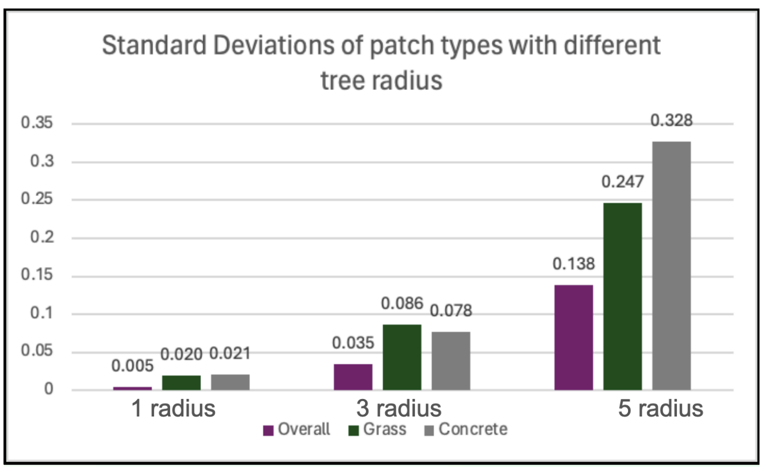


The darker areas are concrete, the middle red is grass, and the lightest is the shade from trees. As you can see, most of the shade covers grass and only a little reaches the concrete. If we changed the strategic trees mechanism to place trees closer to concrete then we would likely see more pronounced effects on the concrete patches that lead to a greater standard deviation than we see currently. Right now most of the effect is concentrated on grass patches, and in order to most effectively alleviate urban heat, shading concrete is more effective since it holds onto more heat.

Tree Radius Scenario Testing

Changing Parameters			
Concrete	Clustering	Number Trees	Radius
50	10	10	1 / 3 / 5

Boston	Overall Mean (30 Runs each)	Grass Mean (30 runs each)	Concrete Mean (30 runs each)
1 Radius	90.793	78.67	102.917
3 Radius	90.118	77.59	102.64
5 Radius	88.74	75.896	101.58

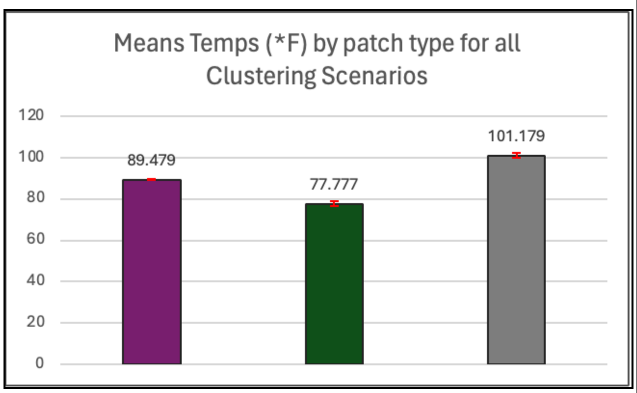


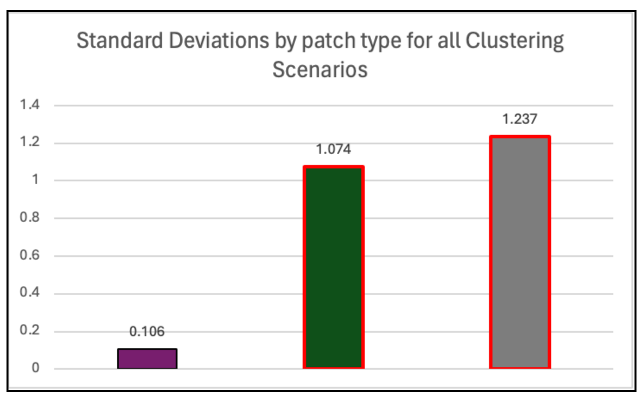
Now exploring how the size of tree shading affects our model, there were some pretty interesting conclusions. The low standard deviation in temperature when tree radius is 1 makes a lot of sense because there is very little shade cooling any of the patches. Going up to radius of 3, we start to see more variation among the different patches. Grass patches still have the most, which makes sense given the tree placement mechanism mentioned earlier and how the effects are greatest on grass. The SD for concrete also increases, though it is not quite at the level of grass. Lastly, looking at a radius of 5, we see the most interesting result, the SD of concrete being the highest. This is probably due to the radius now at least matching the maximum distance that trees can be from concrete, and so much more of the radii of the trees in our model actually overlaps with concrete, giving us the drastically increased SD that we see above.

Clustering Scenario Testing

Changing Parameters			
Concrete	Clustering	Number Trees	Radius
50	0/5/10	10	4

Boston	Overall Mean (30 runs each)	Grass Mean (30 runs each)	Concrete Mean (30 runs each)
0 Clustering	89.365	**79.273**	**99.456**
5 Clustering	89.503	**77.145**	**101.86**
10 Clustering	89.569	**76.913**	**102.22**





Lastly, looking at how the clustering affects our model, we found some remarkably interesting results that are arguably the most important from our testing. As clustering increases, grass temperatures decrease, but concrete temperatures increase. The overall stays more or less the same. What this shows us is that when different landcovers are closely intertwined, the difference between them is smaller than when they are more segregated. This is especially important for the concrete because the concrete is what reaches dangerous temperature levels, and so having the grass be more intertwined with the concrete can reduce the dangerous temperatures. These findings are also supported by the standard deviations, which show greater variation among grass and concrete patches compared to the overall of the model.

Overall Scenario Testing Findings

Concrete/Quantity behaved somewhat how we expected, with the overall temperature of the model changing when the quantity of concrete changed as well. This is not a new finding, and has been supported by existing research on urban heat. It's the same story with the tree testing, with more trees leading to lower temperatures in ways that we expected to see. The effect was more pronounced on grass patches, but that makes sense given that grass patches were receiving the most shade. This is one area we could change our model in the future and explore is the tree placement and how close trees are to concrete, along with the radius. Digging into the radius more specifically, we found that more shading, as expected, led to more cooling. It was interesting to see the high SD for concrete with tree radius of 5, but again that's related to the placement mechanism and is something we can change in the future to test further without the hardcoding of tree placement limits.

Then, there is clustering. Our testing of clustering revealed arguably the most interesting and impactful information about the dynamics of urban heat. If grass and concrete are more mixed rather than segregated, then the temperature of concrete is brought down, and since concrete temps are the main drivers of the "excess heat" that we see in urban areas in our model, these lower concrete temps are in line with a finding that we'd deem "useful" for policymakers and researchers for addressing the problem of urban heat.

Policy/Research Implications

Now comes an important piece of our model; so what? Why does this matter? Why is ABM the right tool for understanding urban heat? What are the potential implications of this tool? Thinking back to Oke (1982), it's explained that observing urban heat in cities is expensive and difficult, mostly due to logistical hurdles like time, amount of space to be sampled, etc. Therefore, our ABM of urban heat (and others like it) can fill the gap and provide insights that, while not actual measurements of real-world conditions, give clarity to the dynamics and interactions of urban environments and how they affect temperature. This is useful for policymakers and researchers who may be attempting to address this problem, but lack the understanding of the dynamics of urban heat.

Using a couple of results from running our model, even with just the basic parameters to begin the scenario, we can see that quantity has a major effect on the overall temperature of a given area. This was already well known however, so it's not really any significant insight that policymakers or researchers did not know. The advice to people in these positions, from these initial insights though, would be to increase grass and decrease concrete wherever possible.

To attempt to build a more interesting and insightful picture for policymakers and researchers, we tested parameters other than concrete to understand the spatial dynamics of the different landcovers and how they affect the temperatures. We found that when concrete and grass are more mixed rather than segregated, concrete can cool down by a few degrees, which reduces the severity of urban heat and thus the problem as a whole. Therefore, one major policy recommendation we can make is for there to be more effort towards mixing grass and concrete as best as possible.

Thinking about this in the Boston context, we know that some trains on the green line are between the roads going different directions, and they have impervious surfaces which increase heat. Therefore, we'd like policymakers to consider options like those in Europe, where some street cars go over grass. This is one example of a way to increase grass in an urban area without causing major changes to the current infrastructure.

Additionally, implementing clever stormwater management can be a way to increase the amount of green in an urban area. This is relevant because effective stormwater management usually involves replacing sections of road or sidewalk (concrete) with plants and greenery, which helps absorb excess water in urban areas but could also lead the areas to be cooler if they are reducing the overall amount of concrete.

Future Work

We could extend our model by integrating social behaviors that capture how people respond to high temperatures, especially using data on populations that are vulnerable to heat. Our model can also be generalized beyond one city by adjusting minimum and maximum daily temperatures to reflect different climates and seasonal conditions around the world, also adding another possible layer of analysis. In addition, adding dynamic tree growth would allow trees to gradually increase in height and canopy size, creating more realistic shading and long-term cooling effects.

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Appendices (Pseudo-Code)

1. **Declare variables**
   1.1. PATCHES:
      1.1.1. Airtemp. each patch stores its own air temperature
      1.1.2. Landcover. each patch stores its own landcover value
      1.1.3. Landcover-type. denotes the type of landcover the patch is
      1.1.4. Has-tree. marks whether a patch contains a tree
      1.1.5. Near-tree. Attribute that stores if the patch is within a radius of at least one tree
   1.2. GLOBALS**:**
      1.2.1. Concrete-temp. Mean air temperature of all concrete patches
      1.2.2. Grass-temp. mean air temperature of all grass patches
      1.2.3. Base-temp. baseline temperature determined by time of day
      1.2.4. Grass-for-tree-patches. Determines what patches are eligible for trees
2. **Setup**
   2.1. Clear all variables
   2.2. Call setup-land
      2.2.1. If random-layout? is turned ON:
         2.2.1.1. Ask patches:
            2.2.1.1.1. Generate random number between 0–100 and store as landcover value
         2.2.1.2. Repeat Grouping X times (determined by slider)
            2.2.1.2.1. Diffuse the landcover values across the patches to create blobs
         2.2.1.3. Count total number of patches
         2.2.1.4. Calculate how many should be concrete
            2.2.1.4.1. Concrete-patches = (concrete slider / 100) * total patches
         2.2.1.5. Set a concrete counter to 0
         2.2.1.6. Sort the patches by landcover value
         2.2.1.7. For each patch in the sorted list
            2.2.1.7.1. If concrete counter is less than concrete patches
               2.2.1.7.1.1. Set the landcover-type to concrete
               2.2.1.7.1.2. Set airtemp to base-temp
               2.2.1.7.1.3. Increase concrete counter by 1
            2.2.1.7.2. Else
               2.2.1.7.2.1. Set the landcover-type to grass
               2.2.1.7.2.2. Set airtemp to base-temp
      2.2.2. If random-layout? is turned OFF:
         2.2.2.1. Ask patches
            2.2.2.1.1. If pxcor is less than or equal to Concrete (SLIDER)
               2.2.2.1.1.1. Set landcover-type to Concrete
            2.2.2.1.2. If pxcor is greater than Concrete (SLIDER)

2.2.2.1.2.1. Set landcover-type to Grass
2.2.2.1.3. Set airtemp to temp-inflow
2.3. Call setup-color
2.3.1. Ask patches:
2.3.1.1. If landcover-type is Concrete
2.3.1.1.1. Set patch color gray
2.3.1.2. Otherwise (If landcover-type is NOT Concrete):
2.3.1.2.1. Set patch color green
2.4. Call setup-trees
2.4.1. Define tree-patches as all eligible grass patches using eligible-tree-patches reporter
2.4.1.1. Set grass-for-trees-patches as all patches with landcover type "Grass"
2.4.1.2. If strategic-trees is turned on
2.4.1.2.1. Filter grass-for-trees-patches so that only grass patches within a radius of 5 patches from any concrete patch remain
2.4.1.3. Report grass-for-tree-patches
2.4.2. Randomly select a number of patches equal to the num-trees slider
2.4.2.1. For each selected patch mark the patch as having a tree
2.4.3. For every patch that has a tree
2.4.3.1. Change its color to a darker green (green - 1)
3. Go
3.1. Call update temp inflow
3.1.1. Make sure each tick is representative of 1 hour of the day, on scale of 1-24
3.1.2. If it is earlier than hour 6
3.1.2.1. Set temperature inflow to be the minimum temp
3.1.3. If it is hour 6 or earlier than hour 14
3.1.3.1. Take minimum value from slider, and take current hour and subtract 6, then multiply that result by difference between (max and min) divided by 8
3.1.4. If it is hour 14 or earlier than hour 18
3.1.4.1. Take maximum value from slider, and take current hour and subtract 14, then multiply that result by difference between (max and decline) divided by 4
3.1.5. If it is hour 18 or later
3.1.5.1. Take decline value from slider, and take current hour and subtract 18, then multiply that result by difference between (decline and min) divided by 6
3.2. Call update tree proximity
3.2.1. Assign patches as near a tree if they are in the radius of a tree
3.3. Call change airtemp

- 3.3.1. Have every patch
  - 3.3.1.1. Apply inflow
    - 3.3.1.1.1. Take the base temp and multiply by inflow rate, applying heat to the patches
    - 3.3.1.1.2. If near a tree
      - 3.3.1.1.2.1. Set inflow to inflow times 1 - tree inflow, reducing the amount of heat entering under trees
    - 3.3.1.1.3. Set the air temperature to include inflow of patches
  - 3.3.1.2. Apply outflow
    - 3.3.1.2.1. Make outflow rate start at 0
    - 3.3.1.2.2. Set outflow rate to concrete rate for concrete patches
    - 3.3.1.2.3. Set outflow rate to grass rate for grass patches
    - 3.3.1.2.4. If the patch is under a tree
      - 3.3.1.2.4.1. Make the outflow rate 1 times tree outflow
    - 3.3.1.2.5. Apply outflow mechanism
    - 3.3.1.2.6. Diffuse amount of temp based on diffuse slider

3.4. Call color heat
- 3.4.1. Local low temp variable is set to min slider - 50
- 3.4.2. Local high temp variable is set to max slider + 50
- 3.4.3. Ask patches
  - 3.4.3.1. Set the colorscale to red based on the airtemp, with low temp being coolest color (lighter red) and high temp being warmest color (darker red)

3.5. Define the mean of patches that are concrete

3.6. Define the mean of patches that are grass

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